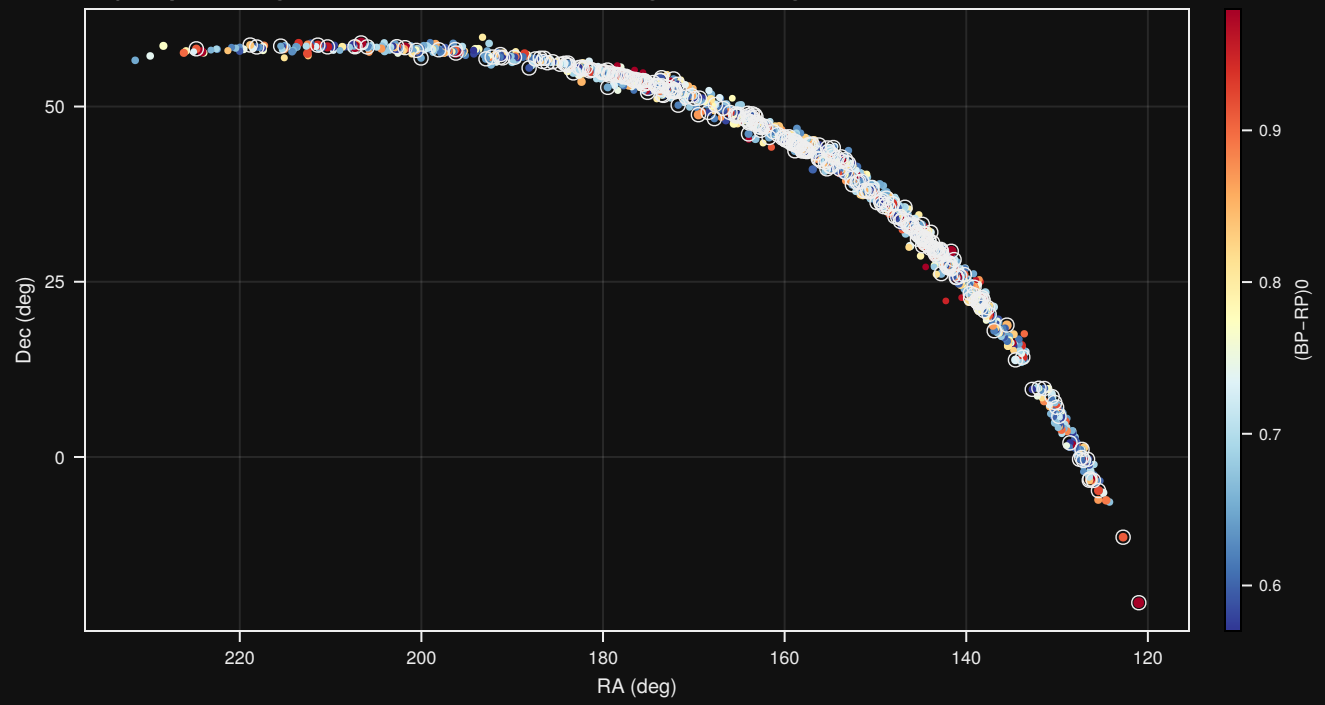
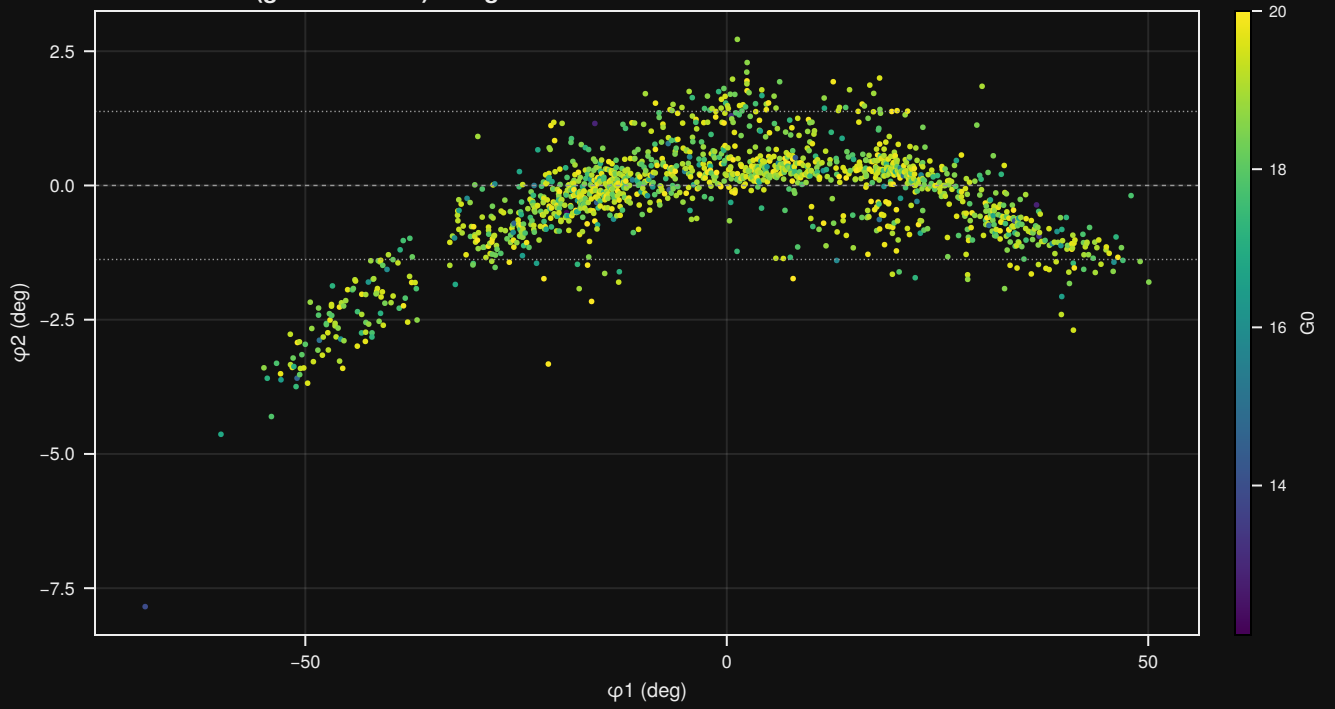
GD-1

MEMBERSHIP	
Stars in catalog	1468
with Gaia GSP-Phot	703 (48%)
with radial velocity	323
Stream id (sID)	53
GEOMETRY · great-circle fit	
Centroid RA, Dec	159.4, 41.5 deg
Centroid l, b	173.2, 54.6 deg
Pole RA, Dec	34.0, 30.4 deg
Length (φ1, 2–98%)	89.4 deg
Width σ(φ2), robust	0.69 deg
KINEMATICS	
pmRA* med ± σ	-6.09 ± 2.10 mas/yr
pmDE med ± σ	-11.29 ± 2.42 mas/yr
μφ1* med ± σ	-12.31 ± 1.55 mas/yr
μφ2 med ± σ	-3.24 ± 0.75 mas/yr
Parallax (cat.) med	0.124 ± 0.221 mas
Parallax (wtd, ZP-corr.)	0.133 ± 0.010 mas
Distance	d = 7.5 ± 0.6 kpc
Physical length	11.74 kpc
RV median ± σ(MAD)	-82.8 ± 146.5 km/s
RV gradient dV/dφ1	-6.43 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.1–20.0 (19.0)
(BP−RP)0 median	0.68
GSP-Phot [M/H] (n=703)	-1.97 ± 0.78
GSP-Phot Teff median	6091 K
RUWE > 1.4	0.0 %
PM clusters (largest)	84 (33 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	1468 | 1468
n_RV catalog | paper	323 | 323
<[Fe/H]> paper	-2.49
Stellar mass (paper)	14000 Msun
Pole offset vs paper	0.9 deg

Sky map · catalog members (n = 1468) · size ∝ brightness · ring = has RV

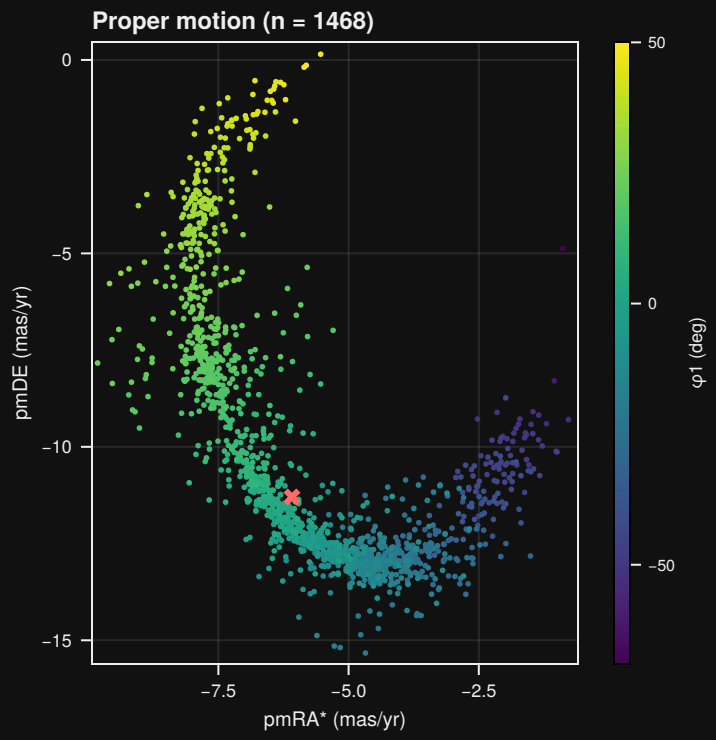


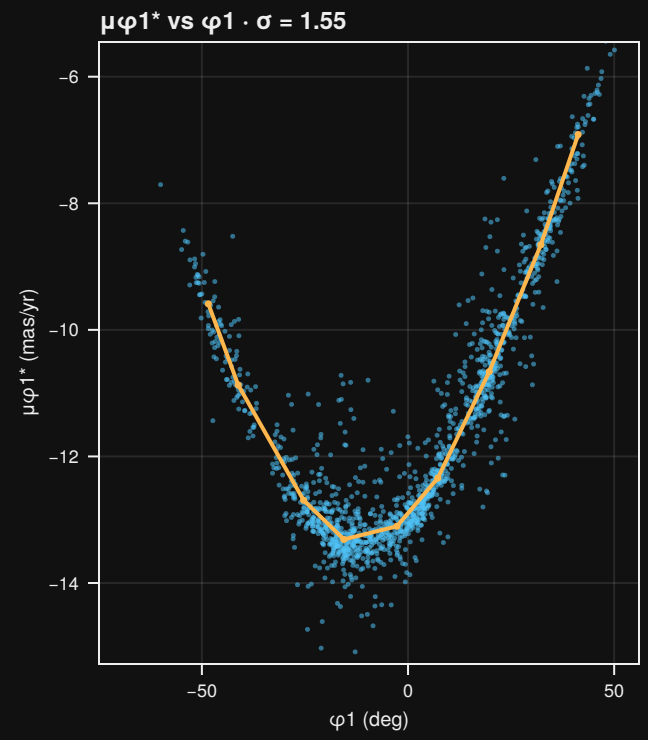
Stream frame (great-circle fit) · length 89.4° · width σ 0.69°

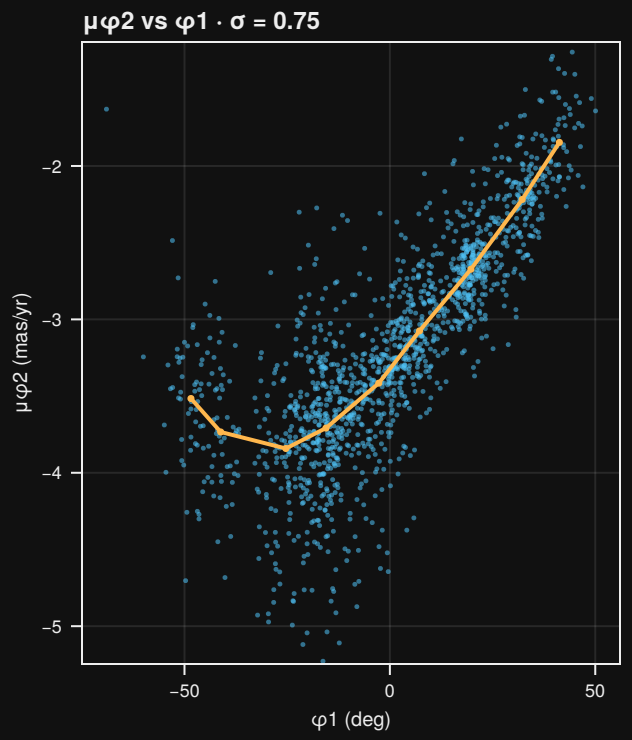


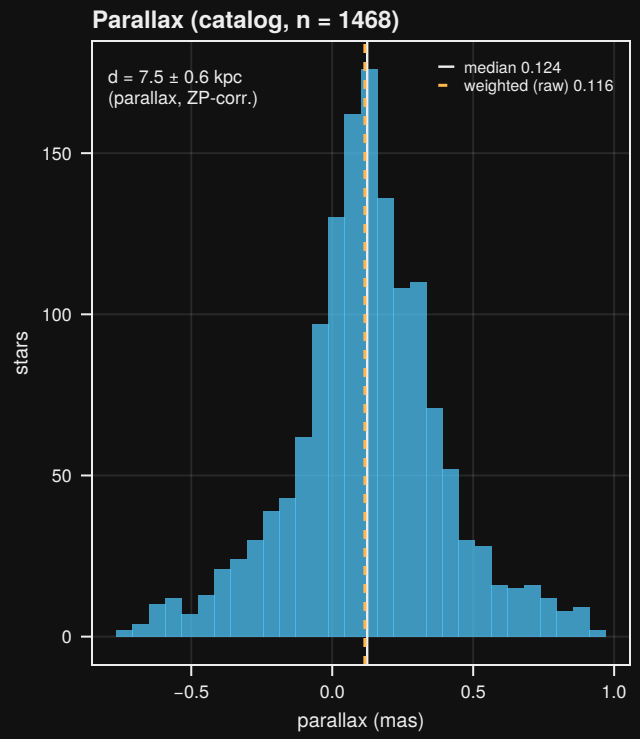
GD-1

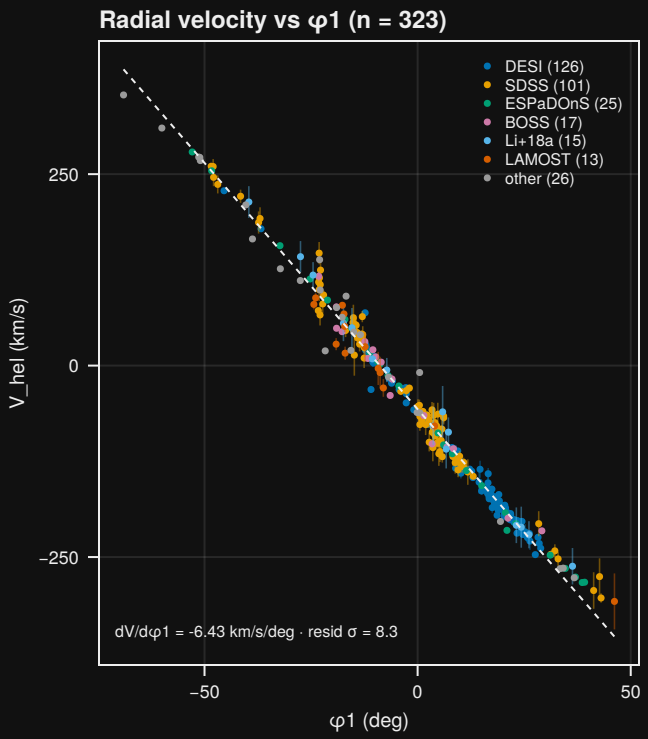
Kinematics · proper motion · parallax · radial velocity

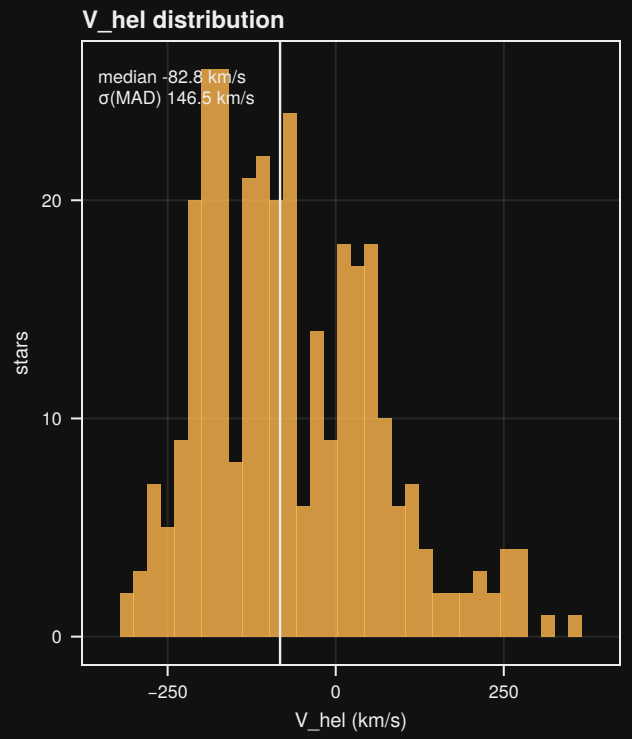






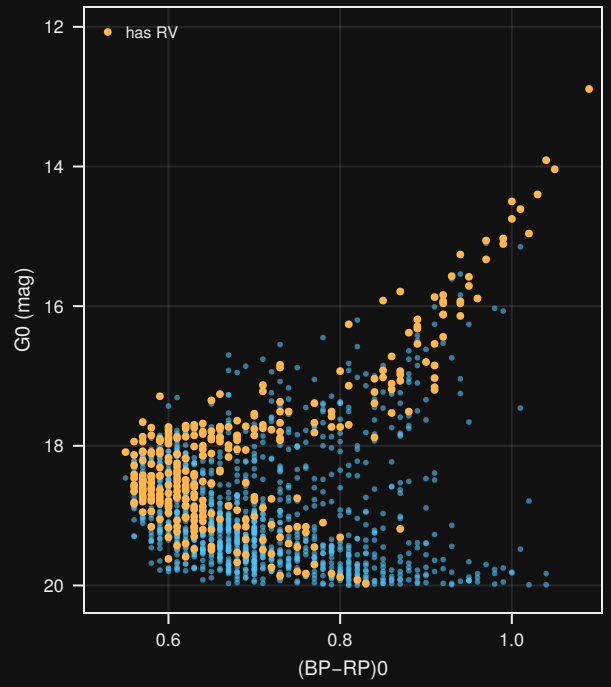




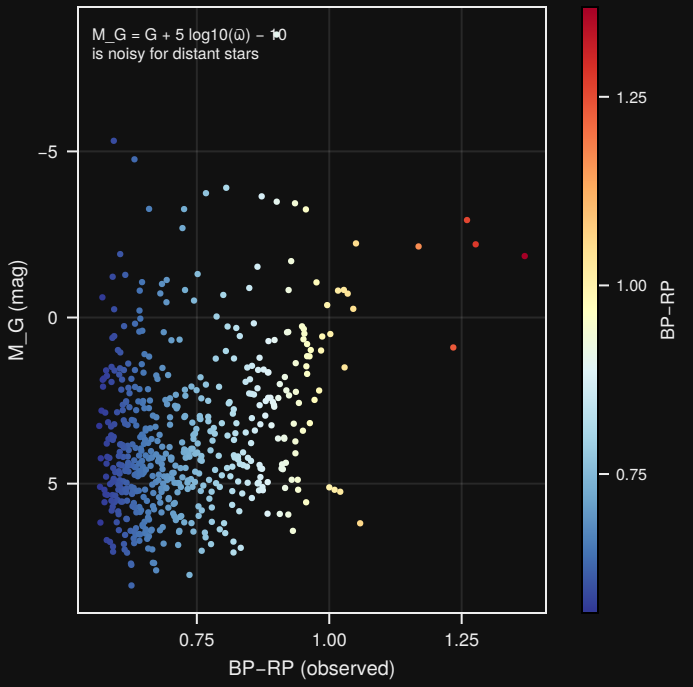


GD-1

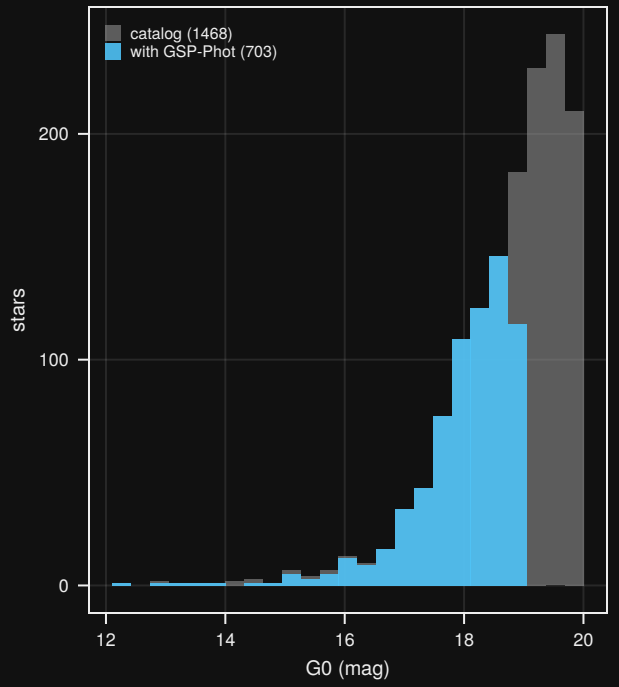
CMD · dereddened (n = 1468)



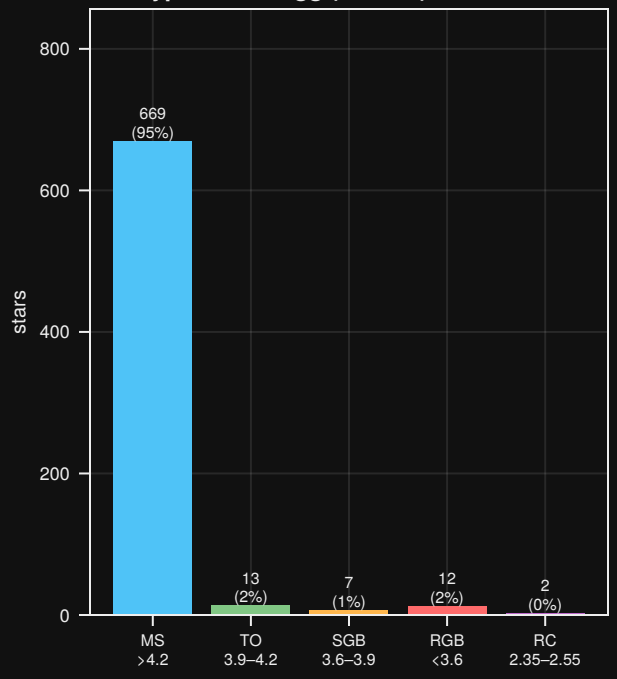
HR diagram · ϖ>0 (n = 555/703)



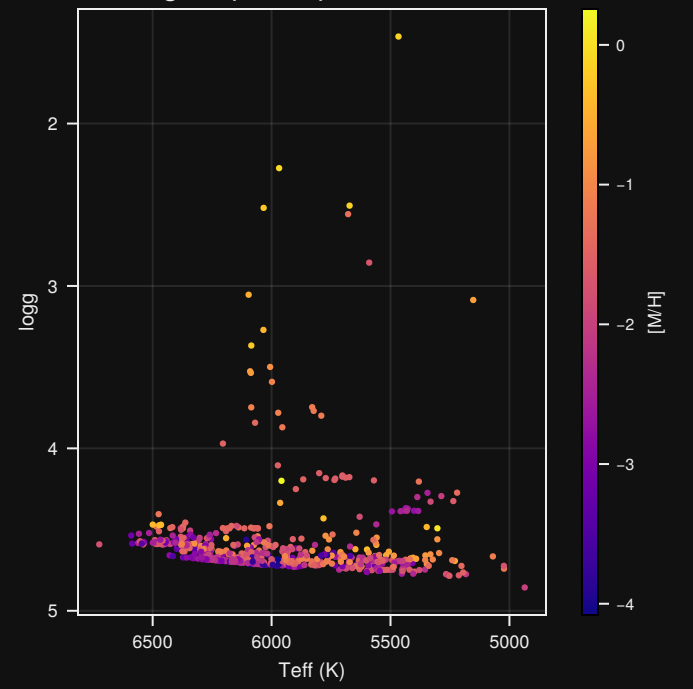
G0 & GSP-Phot coverage



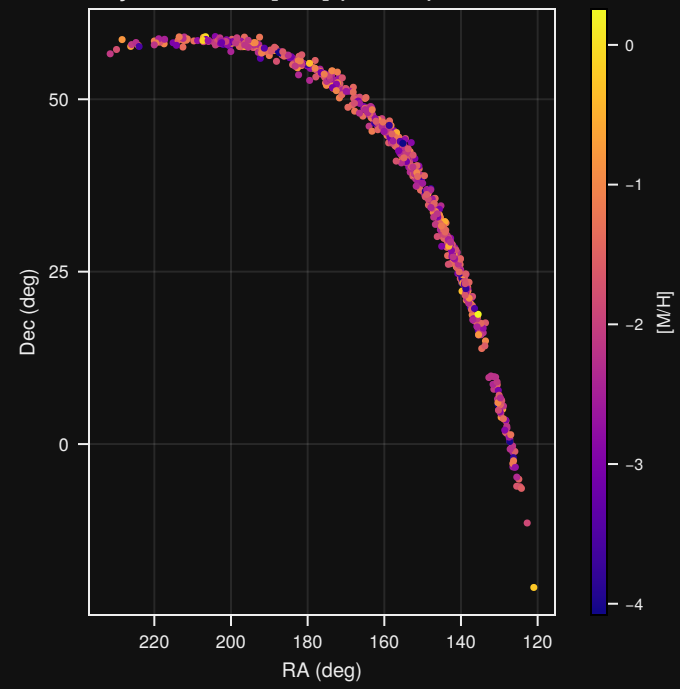
Star types from logg (n = 703)



Kiel diagram (n = 703)



Sky · GSP-Phot [M/H] (n = 703)

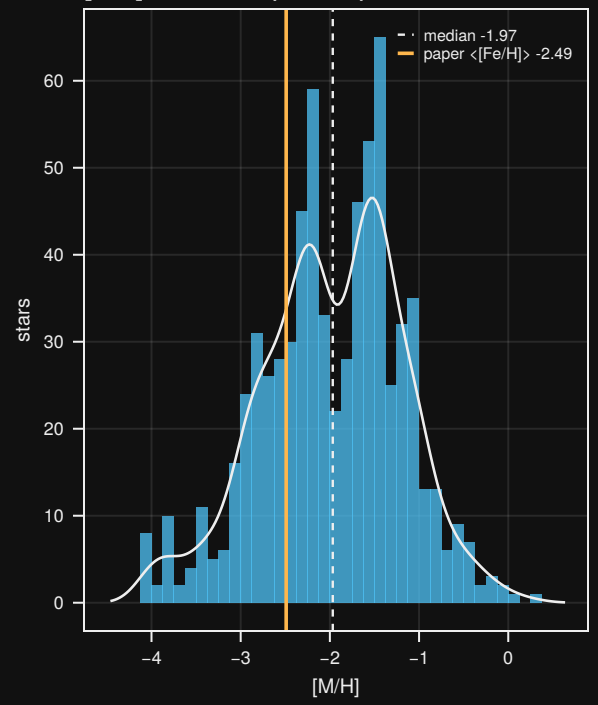


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

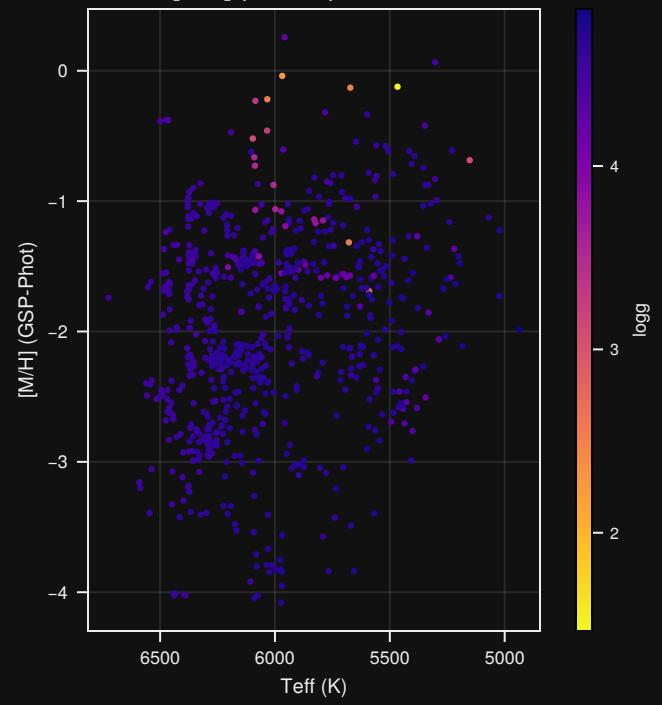
GD-1

Metallicity · age · astrometric quality

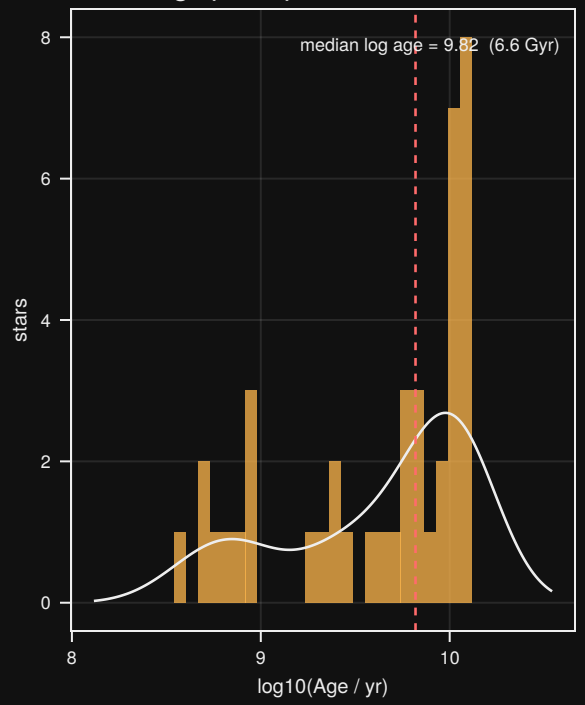
[M/H] GSP-Phot (n = 703)



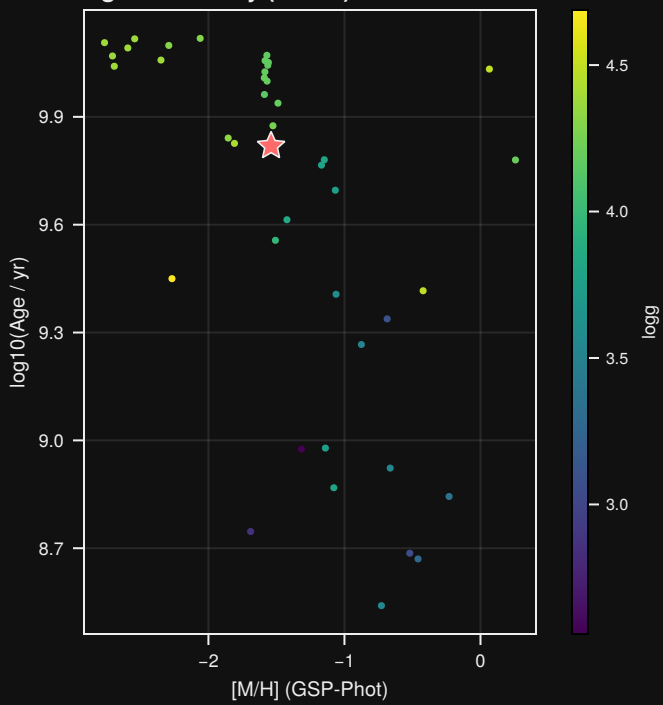
Teff vs [M/H] (n = 703)



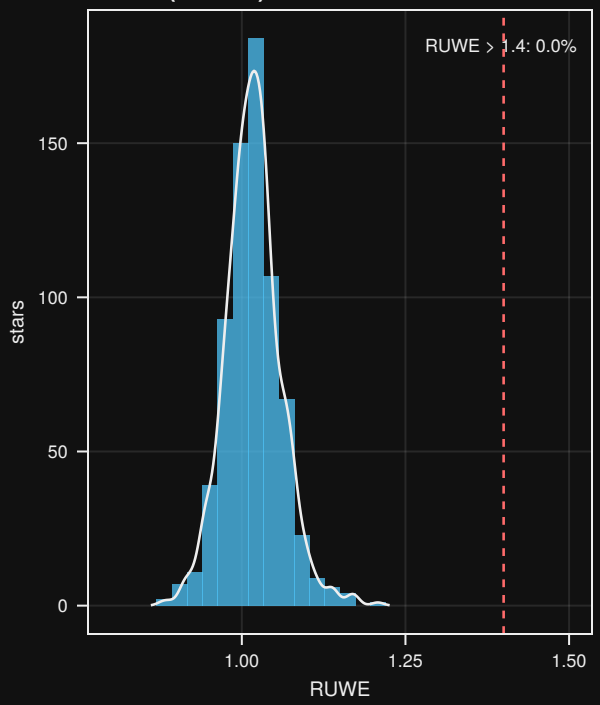
FLAME age (n = 42)



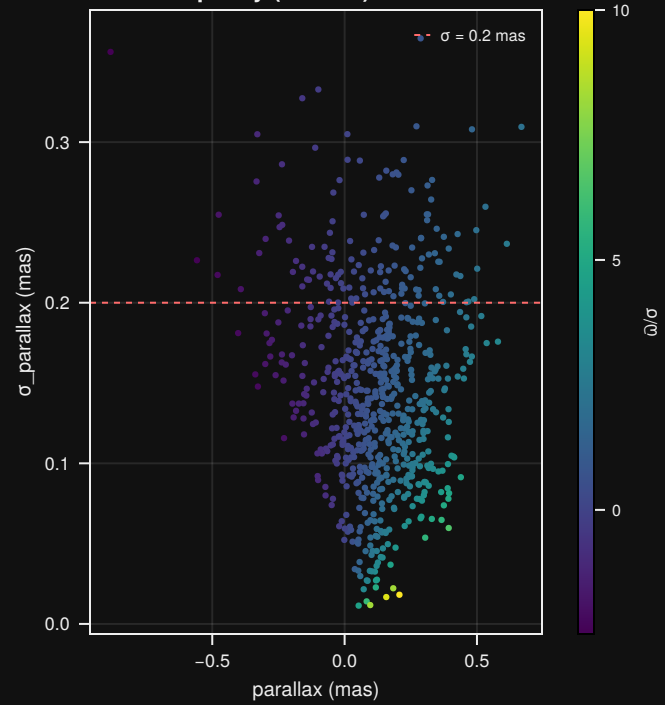
Age–metallicity (n = 42)



RUWE (n = 703)

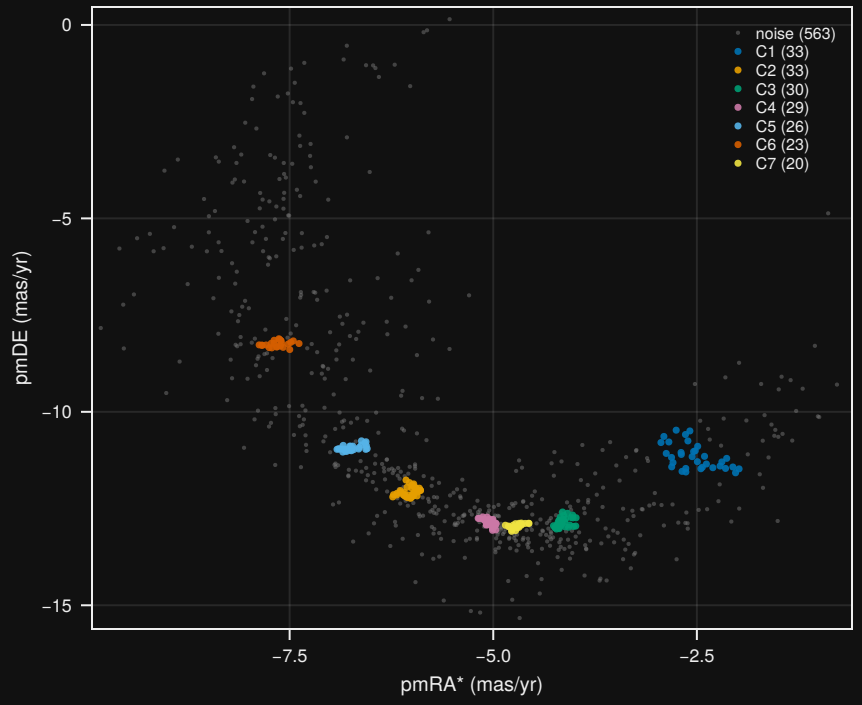


Parallax quality (n = 703)

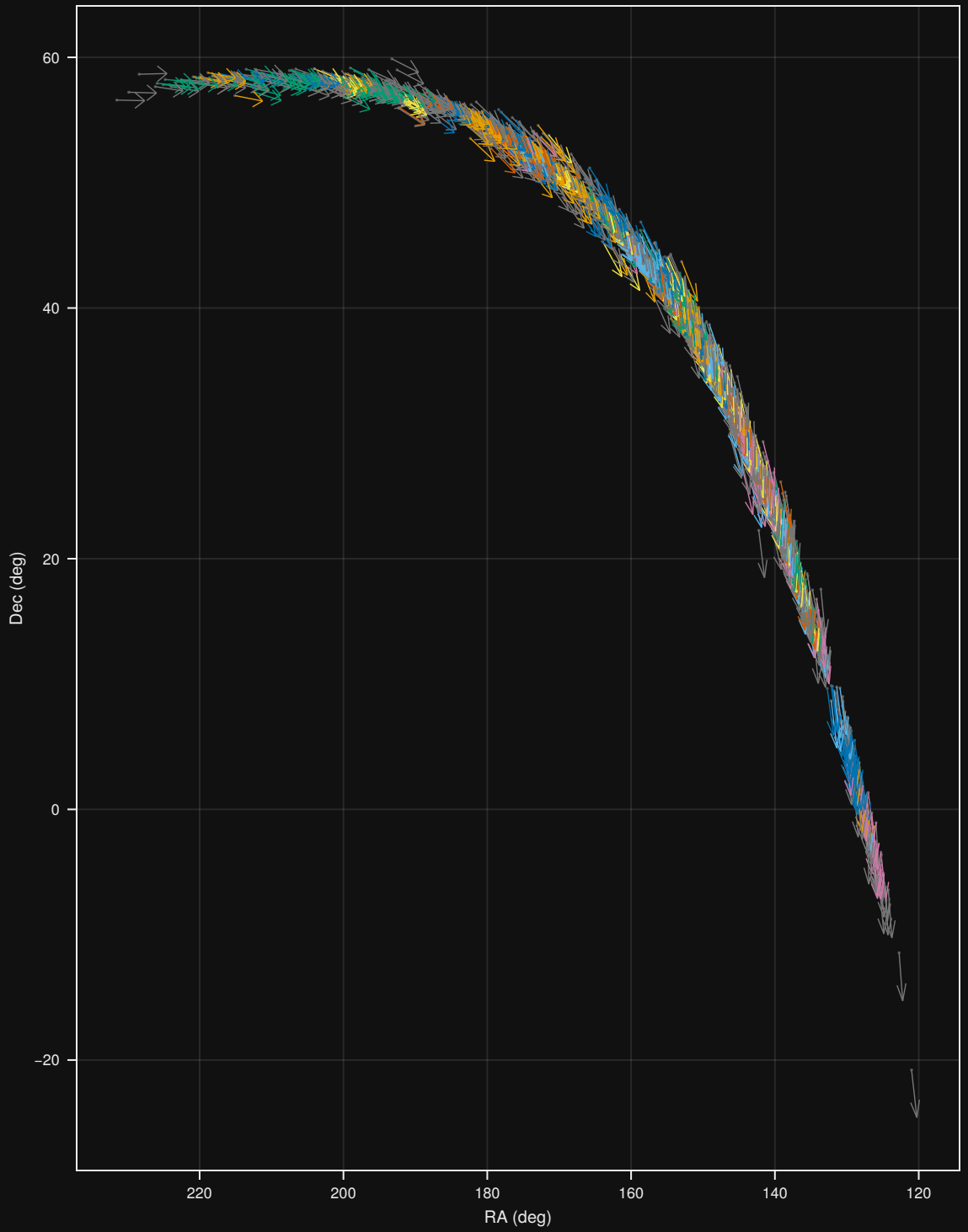


GD-1

PM clustering · HDBSCAN · 84 cluster(s) · noise 38%



Proper-motion directions (arrows) coloured by cluster



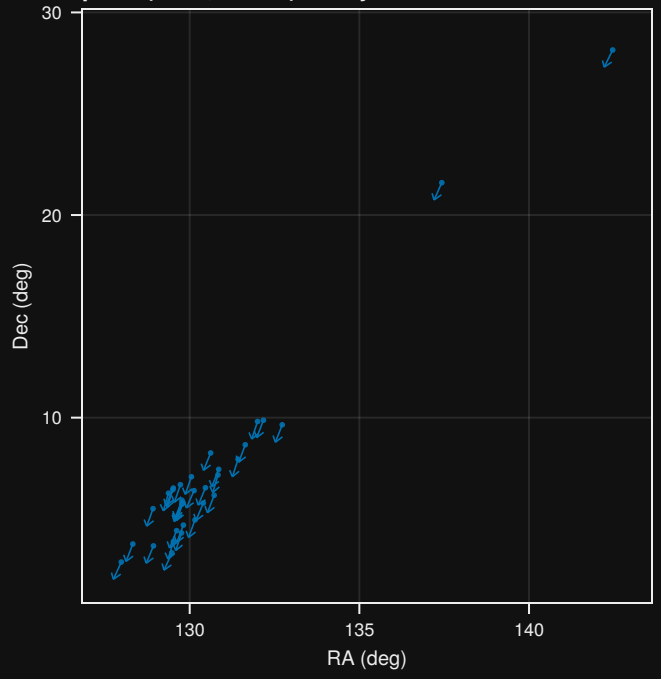
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	563	-5.86	-11.41	2.15	-1.67	87
C1	33	-2.56	-11.20	0.26	-2.18	4
C2	33	-6.02	-12.08	0.08	-1.72	8
C3	30	-4.15	-12.90	0.07	-2.47	6
C4	29	-5.03	-12.88	0.07	-1.96	9
C5	26	-6.75	-10.96	0.15	-1.99	10
C6	23	-7.63	-8.25	0.13	-2.20	7
C7	20	-4.70	-12.93	0.07	-2.19	6
C8	20	-7.83	-7.59	0.09	-1.67	11

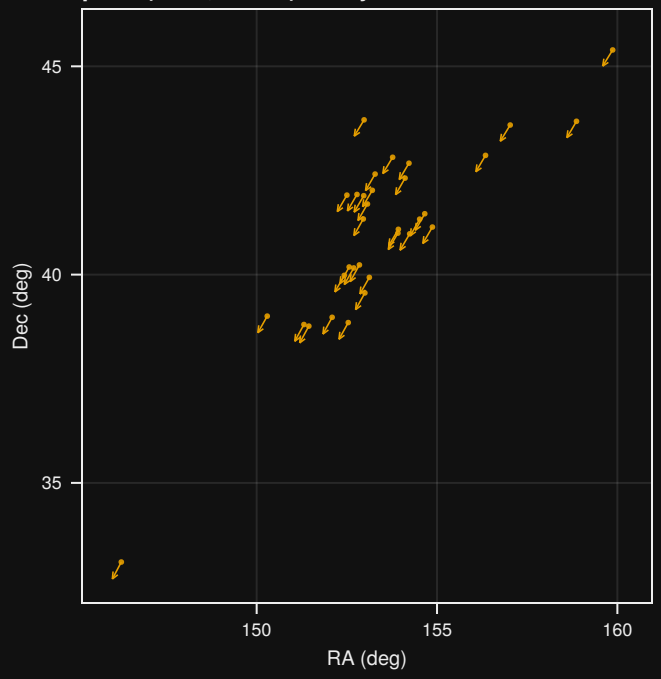
GD-1

Per-cluster proper-motion vector maps

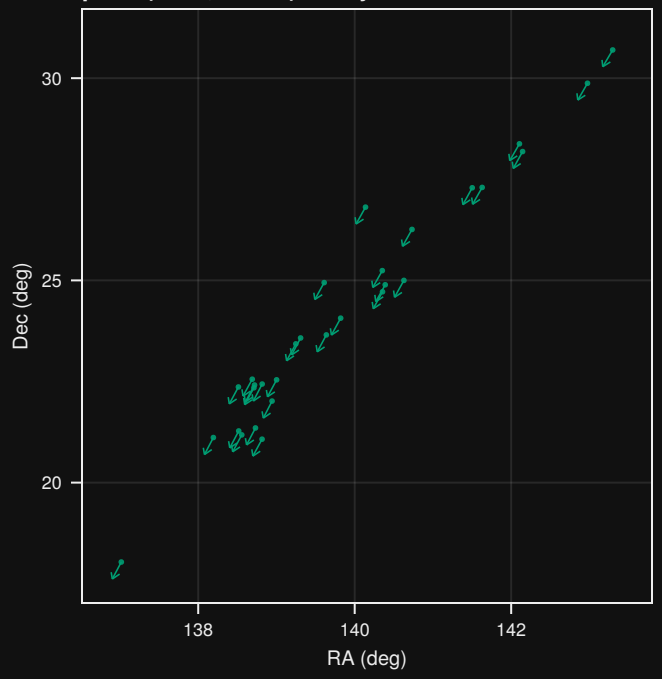
Cluster C1 · n = 33
pm = (-2.56, -11.20) mas/yr



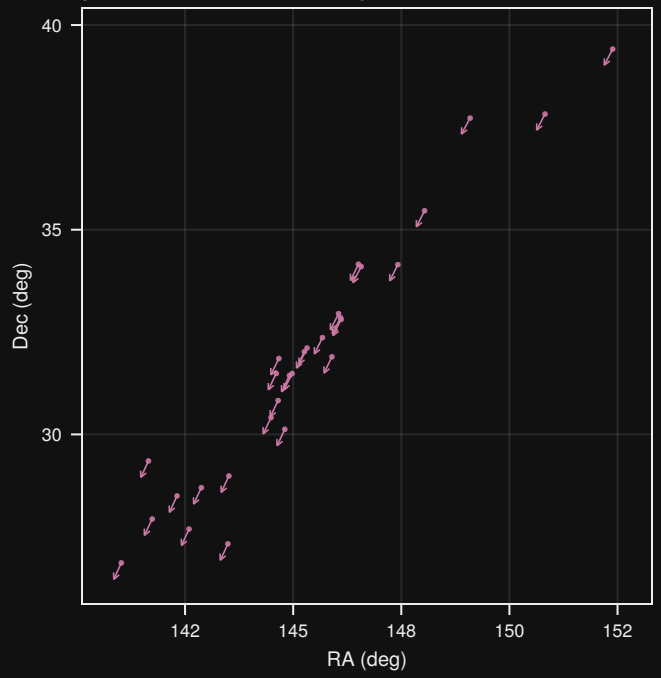
Cluster C2 · n = 33
pm = (-6.02, -12.08) mas/yr



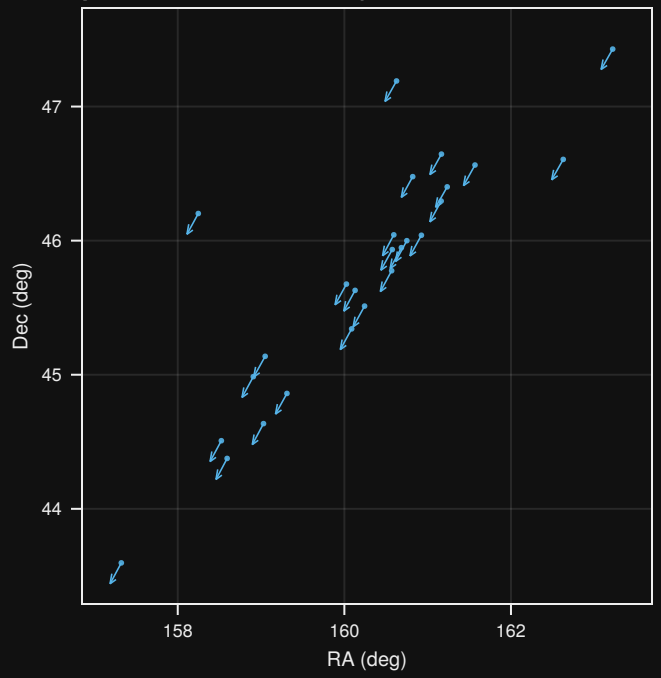
Cluster C3 · n = 30
pm = (-4.15, -12.90) mas/yr



Cluster C4 · n = 29
pm = (-5.03, -12.88) mas/yr



Cluster C5 · n = 26
pm = (-6.75, -10.96) mas/yr



Noise · n = 563

